Helium Step Heats on Geothite at Water Saturated Conditions Show Diffusive Helium Loss and High Retentivity at Surface Temperatures

Ben J. Thyer[a,*], Jonathan Treffkorn[a], Kenneth A. Farley[a]

[a] *Division of Geological and Planetary Sciences, California Institute of Technology, Pasadena, 91125, CA, USA*

Abstract

(U–Th)/He dating of goethite (FeOOH) can directly date weathering, ferruginization, and laterite formation, but interpreting goethite He dates requires knowing the conditions under which goethite retains radiogenic helium. Vacuum step heats, the standard tool for characterizing helium retention, convert goethite to hematite and record dehydroxylation rather than diffusion (Farley et al., 2024). Because high water pressure suppresses dehydroxylation, we use the method described in (Thyer et al., 2026) to conduct the first helium step heat experiment on proton-irradiated goethite under water-saturated conditions. Goethite sample Win-06-01b (Flinders Ranges, South Australia) was encapsulated in a silica glass ampoule with distilled water and heated through multiple prograde and retrograde cycles between 170 and 250 °C over about 40 days. Apparent ^{3}He diffusivities are approximately seven natural log units lower than those measured on the same sample in vacuum and, beyond 7% cumulative release, define a linear Arrhenius array with an activa-

*Corresponding author.
Email address: benjthyer@gmail.com (Ben J. Thyer)

tion energy of 157.1 ± 5.3 kJ/mol and a closure temperature of 98.1 ± 8.6 °C at 10 °C/Ma. Reanalysis of two published vacuum step heats of the same sample shows that vacuum release follows first-order conversion to hematite, whereas the water-saturated release shows the history dependence of diffusion and follows multiple diffusion domain systematics. Thermodynamic and nucleation modeling shows that the small driving force for dehydroxylation at water saturation, compared with the unbounded driving force at the near-zero water pressure of a vacuum line, raises the barrier to hematite nucleation enough to suppress solid-state conversion on experimental timescales. Fluid-mediated conversion, observed in synthetic goethite pigment, and goethite recrystallization cannot be excluded; if either contributed, the diffusivities are upper bounds. A four-domain model places 97% of the ^{3}He in a single domain that is nearly quantitatively retentive at 25 °C over billion-year timescales. The ^{4}He/^{3}He spectrum nonetheless requires natural helium loss from this domain, which thermally activated diffusion can explain: steady cooling from 76–80 °C at 150–175 Ma fits best, while burial of surface-formed goethite to a Late Cretaceous peak of 58–65 °C, consistent with regional apatite fission-track data, also fits the ^{4}He/^{3}He spectrum. Water-saturated step heats enable systematic experiments on how crystallite size, composition, and radiation damage control helium retention in goethite.

Keywords: goethite, (U–Th)/He geochronology, helium diffusion, step heating, dehydroxylation, multiple diffusion domains

---

1. Introduction

Goethite (FeOOH) crystallization is datable only by U-Th/He geochronology (Shuster et al., 2005; Farley et al., 2024). Goethite is a powerful chronometer for dating surface processes such as weathering and ferruginization directly: laterite and duricrust formation and recrystallization (Monteiro et al., 2014; Allard et al., 2018; Ansart et al., 2022), and the ferruginization of organic matter and paleosols (Heim et al., 2006; Yapp and Shuster, 2011; Vasconcelos et al., 2013; Danišík et al., 2013; Hofmann et al., 2017).

Interpreting a goethite U-Th/He date requires knowing the conditions under which the mineral retains radiogenic helium. As with any radiochronometer, meaningful geological interpretation of a measured age depends on knowing when the system is open to loss or gain of parent or daughter nuclides. For goethite, step heats of proton-irradiated, crystalline samples are used to reveal the degree of partial helium loss by the $^4He/^3He$ method (Shuster et al., 2005), and suggest that helium loss fractions in nature are always non-zero but modest, ranging from 2 to 30% (Shuster et al., 2005; Heim et al., 2006; Hofmann et al., 2017; Deng et al., 2017). Although one study attempts to explain the variable loss principally as a result of variable radiation damage (Bassal et al., 2022), the effects of temperature, crystallite size, and chemical composition have not been determined experimentally and remain unresolved, because goethite decomposes to hematite (Fe_2O_3) in the step heat experiments used to characterize it (Farley et al., 2024).

Laboratory experiments of helium migration under controlled conditions have proven extremely useful in understanding the mechanisms and rates of helium migration in geological materials and have produced sensible results

that are predominantly explainable by thermally activated volume diffusion (Reiners and Farley, 1999; Farley, 2000, 2007; Cherniak and Watson, 2011; Farley, 2018), albeit variably complicated by the effects of radiation damage and annealing (Shuster et al., 2006; Guenthner et al., 2013). This approach has also been leveraged for goethite; however, some complications arise when applying this approach to goethite. Above surface temperatures, goethite decomposes to hematite and water, often along crystallite elongation directions (Goss, 1987). Farley et al. (2024) measured evolved water by capacitance manometry during vacuum step heats and found helium and water release to be tightly correlated even below 200°C, indicating that decomposition, rather than helium migration through an unchanging crystal, governs the release. Vacuum step-heat results for goethite therefore need not describe helium behavior in nature.

Still, existing helium step heat measurements of goethite in vacuum likely remain informative to an extent, as goethite dehydration kinetics are highly influenced by some of the same factors that could be responsible for variable helium retention in natural goethite, like variable grain size (Navrotsky et al., 2008; Guo and Barnard, 2011; Goss, 1987), crystallinity (Schwertmann et al., 1985) or substitution of aluminum or other cations within the goethite structure (Majzlan et al., 2003; Bassal et al., 2022). For example, Shuster et al. (2005) initially conducted vacuum step-heating experiments on two vitreous, polycrystalline goethite samples, in which uniformly distributed ^{3}He was introduced by proton irradiation for comparison with radiogenic ^{4}He that was partially lost in nature. The results of Shuster et al. (2005) could be well-fit with a model of two distinct diffusion domains or migration mechanisms with

significant differences in helium retention under surface conditions, e.g. crystallite interiors and intergranular space. Although these results may have been impacted by dehydration in vacuum, certain aspects may remain applicable to helium loss in goethite in nature if helium loss in nature and the laboratory advance through the samples with similar geometries. One important point is that if laboratory experiments degas domains in order of their helium retentivity in nature, even if the rates of loss are affected by goethite dehydration, then a $^4He/^3He$ experiment in vacuum can still be used to measure helium loss (Shuster et al., 2005; Vasconcelos et al., 2013).

An additional complexity is that natural goethite almost always occurs as aggregates of minute crystallites with cross-sections smaller than 1 µm in nature. Neglecting to account for multiple diffusion domains can lead to inaccurate determinations of activation energy (E_a) and frequency factor (D_o/a^2) for helium diffusion (Farley, 2018). Detailed step-heating with temperature cycling, as has been done for hematite (Farley and Flowers, 2012; Farley, 2018), is needed to characterize domain size and activation energy for polycrystalline samples.

Goethite dehydration is suppressed at high water pressure (Langmuir, 1971; Navrotsky et al., 2008), so a step heat conducted under water vapor saturated conditions may interrogate helium loss rather than decomposition kinetics. Until now, no studies have reported helium step-heating results for goethite under water vapor saturated conditions, which could help stabilize goethite or slow its decomposition to illuminate the processes relevant to helium loss in natural settings. If experimental conditions can be controlled to slow the decomposition of goethite such that the rate of natural helium loss

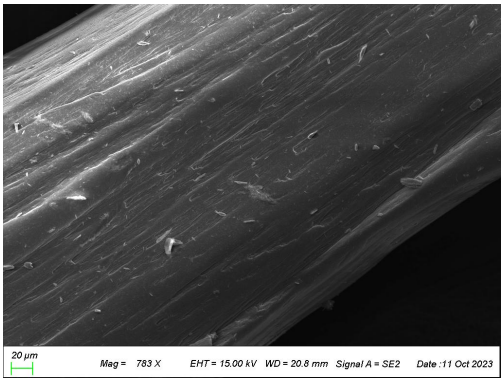
in goethite greatly exceeds the rate of goethite decomposition into hematite and water, it would open up an experimental approach to systematically assess how factors like crystallite size, temperature, radiation damage, and chemical composition could affect helium mobility and retentivity in nature (Bassal et al., 2022).

Early trials of the method were encouraging (§2.3). SEM imagery taken of thermally treated natural goethites in vacuum and open air showed the development of contraction cracks on the outside and throughout the interior of goethite grains heat treated in vacuum and in open air (Fig. 1b,d), while goethite heat-treated when encapsulated with water did not form cracks (Fig. 1c). Goethite heat treated in open air turned red, whereas goethite heat treated in vacuum remained black.

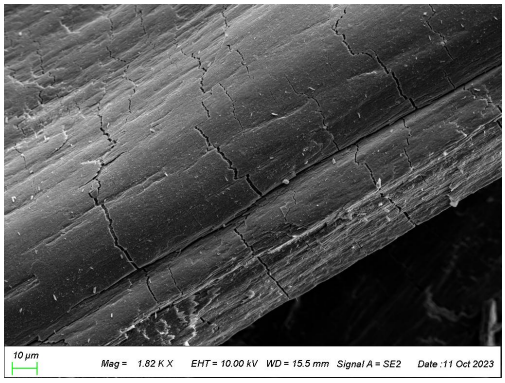
We test this premise with the first helium step heat experiment on goethite under water-saturated conditions, and compare it with published vacuum step heats of the same sample (§4.1–4.2). A thermodynamic and kinetic analysis of goethite dehydroxylation then explains why water has this effect (§4.3).

2. Methods

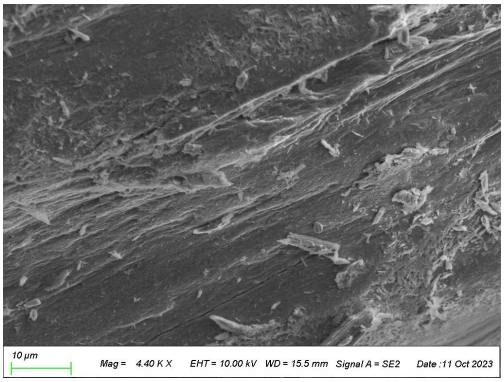
This study builds upon previous helium step heat studies of the dense colloform goethite sample Win-06-01b (occasionally referred to as Win0601b) collected within float at the Flinders Ranges in South Australia (Waltenberg, 2012; Monteiro, 2017). It yields one of the oldest known goethite U-Th/He ages: 180-135 Ma (Waltenberg, 2012). To produce a homogeneous distribution of ^{3}He, Win-06-01b was irradiated with 220 MeV protons for 24 hours



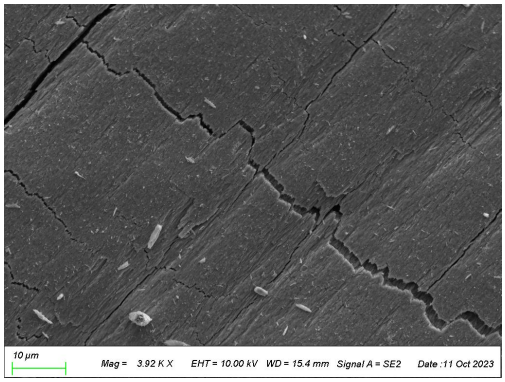
(a) Control



(b) Vacuum, 300°C, 11 h



(c) Water-buffered, 300°C, 4 h



(d) Vacuum, 300°C, 11 h

Figure 1: Secondary electron images of Bahia goethite aggregate before and after heat treatment at 300°C. (a) Untreated control grain; it has some rough surfaces and rare long-axis-parallel cracks. (b) After 11 hours in vacuum, cracks developed both parallel and perpendicular to the long axis of the polycrystalline aggregate, representing volume contraction due to dehydration/decomposition of goethite into hematite. (c) After 4 hours in a sealed glass tube with water, no cracks indicative of dehydration were visible; most of the aggregate's surface was smoother than the area pictured, which had the roughest texture on the grain. A very small crack parallel to the long axis may indicate some dehydration had occurred, or it may have been inherited from before treatment at water vapor saturation, as observed in the control. (d) The vacuum-heated grain of (b) at the same scale as (c), showing rough cracks at various scales both parallel and orthogonal to the long axis; these are the typical decomposition features that would be expected in (c) if significant decomposition had occurred there.

at the Francis H. Burr Proton Therapy Center, Massachusetts General Hospital, following standard procedures (Shuster et al., 2004) and subsequently was subjected to helium step heat measurements in vacuum by Monteiro (2017) and Farley et al. (2024). Although these two studies used subtly different heating schedules, the results of Farley et al. (2024) clearly agree well with Monteiro (2017).

In this study, an aliquot of the same irradiated Win-06-01b goethite used by Monteiro (2017) and Farley et al. (2024) was encapsulated with MilliQ water within a silica glass tube following Thyer et al. (2026). The ampoule retained all of its water through the experiment, including more than a week at 250°C with an internal vapor pressure of about 40 bar (Haar et al., 1984) (SI Fig. S4).

The sealed ampoule, now referred to as HT-Win-06-01b, was placed in a copper foil packet and mounted to a K-type thermocouple within an ultra-high vacuum projector lamp heating cell described in more detail by Farley et al. (1999). Before the experiment began, the ampoule was loaded in the lamp cell, pumped to high vacuum, and was held at 70°C for 4 days to degas atmospheric helium entrained within the ampoule.

HT-Win-06-01b was subjected to multiple prograde-retrograde heating cycles to investigate possible MDD behavior. Step durations varied from 2 hours to about a week; the total experiment was conducted over about 40 days and included experimental steps ranging from 170-250°C. We elected not to heat the sample hotter out of an abundance of caution, because similar ampoules explode above 300°C (Thyer et al., 2026).

Recorded temperatures oscillated by ~1°C around each target. We believe

that the recorded temperatures are accurate: encapsulated Durango apatite heated in the same way reproduces published diffusivities (Thyer et al., 2026).

Helium isotope measurements of evolved gases from water vapor saturated step heats of glass-encapsulated irradiated HT-Win-06-01b goethite were acquired at Caltech on a Thermo Fisher Helix-MC-Plus^{10K} mass spectrometer in peak hopping mode from January 31st, 2024 to March 13th, 2024. Helium was purified by exposing evolved gases to a charcoal trap held in liquid nitrogen, a titanium sponge getter, then a SAES NP10 getter before freely expanding into the Helix-MC-Plus^{10K}. ^{3}He was measured with a pulse-counting electron multiplier, while ^{4}He was detected on a Faraday detector. Measurements of high ^{3}He MM gas standard were analyzed every 8 hours (totaling 116 measurements) to track instrumental drift of ^{3}He and ^{4}He sensitivities and to quantify internal reproducibility.

A cold blank was measured at the beginning of the experiment, showing a negligible ^{3}He blank and a ^{4}He blank signal corresponding to about 1.25 pccSTP, which was similar to ^{4}He values attained during low temperature permeation step measurements. Permeation steps were thus assumed to measure shifting ^{4}He blanks during the experiment.

For the final step, the HT-Win-06-01b ampoule was broken and dried overnight in a 50°C oven, then completely degassed by heating to ~1100 °C in a resistance furnace. The final step measurement was conducted on a Thermo Fisher Helix SFT mass spectrometer. Blank, re-extract and sensitivity details are given in the SI (§**??**).

Helium released by the sample reaches the mass spectrometer only after permeating the ampoule wall, so each measured step yield lags the true re-

lease. Step yields were corrected for this permeation lag with the bounding method of Thyer et al. (2026), which gives lower-bound, central and upper-bound values for every step. The correction and the ampoule permeabilities measured during the experiment are described in the Supplementary Information (§S1). The propagation of the permeation-lag bounds into apparent diffusivities and isotope ratios are described by Thyer (2026b).

2.1. Kinetic and Diffusive Reductions of Step-Heat Data

We reanalyzed the published vacuum step heats of irradiated Win-06-01b by Farley et al. (2024) (hereafter *KAF*) and Monteiro (2017) (hereafter *HDM*), together with the water-saturated step heat of this study, by reducing every step both as an apparent first-order conversion rate constant and as an apparent diffusivity. For the kinetic reduction, ^{3}He is assumed to be immediately released from the goethite lattice upon conversion to hematite, except for a fraction (ϕ) that is inherited by the hematite product. ϕ itself is a free parameter fit in the least squares optimization, yielding $\phi = 0.101$. This value is consistent with the ^{4}He/^{3}He spectra, which record an abrupt drop at $F_cum \approx 0.9$, corresponding to the onset of the slow, late "Stage 3" helium loss as described by Farley et al. (2024). The kinetic inversions (Eq. (3)) for the latest steps are highly sensitive to the choice of ϕ and are excluded from the fits of the kinetic parameters. The KAF steps run with the liquid nitrogen trap isolated are also excluded from the Arrhenius regression. The fraction of helium released in each step is then $F_i = (1 - \phi)\,\alpha_i$, where α_i is the fraction of goethite converted in step i.

We describe the kinetics of goethite dehydroxylation with a first-order

rate law,

$$\frac{d\alpha}{dt} = k(T)\,(1-\alpha), \qquad k(T) = A\exp\!\left(-\frac{E_a}{RT}\right), \tag{1}$$

For a schedule of isothermal steps of duration Δt_j at T_j, Eq. (1) integrates to

$$\alpha_i = 1 - \exp\left[-A\sum_{j\le i},\Delta t_j \exp\!\left(-\frac{E_a}{RT_j}\right)\right], \tag{2}$$

Inverting Eq. (2) step by step gives an apparent rate constant for every step that is independent of any fitted parameter except ϕ:

$$k_i = -\frac{1}{\Delta t_i}\ln\frac{1-\alpha_i}{1-\alpha_{i-1}}, \qquad \alpha_i = \frac{F_i}{1-\phi}. \tag{3}$$

For comparison, the diffusion parameter of each step is obtained with the same Fechtig and Kalbitzer (1966) inversion for a sphere (Fig. 4). A thermodynamic and kinetic model of goethite dehydroxylation, used in §4.3 to interpret the effect of water in this experiment, is described in the SI (§S3–S4).

2.2. Multiple Diffusion Domain and Thermal History Modeling

Together, the ^{3}He release and the bulk-normalized ^{4}He/^{3}He spectrum constrain the thermal history, if the activation energy for diffusion is known (Farley and Flowers, 2012). We interpret the data from HT-Win-06-01b in two stages. First, the diffusion domain structure of the sample is inverted from the ^{3}He release alone. Second, natural thermal histories are fitted to the ^{4}He/^{3}He spectrum with that domain structure held fixed. Fits for $N = 1$–5 domains and the best-fitting thermal histories are given in the SI (§S5).

Diffusion domain structure. Following Farley and Flowers (2012), the sample is represented by N non-interacting spherical domains that share the activation energy of the Stage 1 regression (§3.1). Each domain holds a fraction of the uniformly distributed, proton-induced ^{3}He at a size relative to the reference domain of the Stage 1 regression (r/r_0), giving $2N-1$ free parameters. These minimize the summed squared misfit of the fractional and cumulative ^{3}He yields, weighted equally, over the 28 steps with more than 0.1 pccSTP of ^{3}He. The gas released in the six low-signal steps still counts toward the cumulative yields of later steps, and their durations toward the heating schedule. Domains are added one at a time, each search starting from the $(N-1)$ domain solution, for $N \in [1, 10]$. The Arrhenius and delta plots are not fitted, but follow from applying the Fechtig and Kalbitzer (1966) inversion to the model cumulative release. Their misfit is reported as a reduced χ^2 over the same 28 steps.

Thermal histories. Given a fixed domain structure, any given thermal history predicts a ^{4}He/^{3}He spectrum and a bulk He age following Shuster and Farley (2004), assuming U and Th are uniformly distributed. We compare two types of thermal histories: steady cooling with two free parameters and burial then exhumation with three free parameters.

- **Steady cooling:** goethite crystallizes at an elevated temperature T_0 at formation age t_{form} (e.g. in a hydrothermal environment) and linearly cools to 25 °C today.

- **Burial and exhumation:** goethite crystallizes at the surface (20°C) at formation age t_{form}. It is buried and linearly heated, reaching a

peak temperature T_{max} at time t_{peak}. It is then exhumed to the surface by the paleogene: it is linearly cooled to (20°C) at 50 Ma. This model roughly follows the thermal history Gibson and Stüwe (2000) and Mitchell et al. (1998) interpreted for the region from apatite fission track thermochronometry.

Each path is scored by the χ^2 misfit of its predicted $^4He/^3He$ spectrum to the observed spectrum over the 24 steps that have more than 0.1 pccSTP of ^3He, taking each uncertainty as half the range between the lower- and upper-bound ratios. The reduced χ^2 uses $\nu = 22$ degrees of freedom for steady cooling and $\nu = 21$ for burial and exhumation. χ^2 is evaluated for these free parameters along regularly spaced grids to produce a likelihood heatmap (Fig. 9). The measured He age range (130–180 Ma; Waltenberg 2012) is treated as a hard constraint. The best fit of each family is the lowest-χ^2 path with a predicted age in that range.

2.3. SEM Study of Thermally Treated Goethite

In early trials of the "glass tube method" for helium step heat experiments with controlled microenvironments presented here, sub-samples of goethite morphologically similar to Win-06-01b from Igarapé Bahia (Shuster et al., 2005; Monteiro, 2017) were heated in vacuum using the projector lamp heating apparatus of Farley et al. (1999) and in sealed glass ampoules with water additionally added to provide moisture to slow/prevent dehydration of goethite. All samples were held at 300°C for 4-11 hours, and the glass ampoules were broken and the goethites were allowed to air dry before all samples were placed on carbon tape and carbon coated. These samples were

imaged at Caltech in secondary electron mode by the ZEISS 1550VP Field Emission SEM.

3. Results

3.1. Water-Saturated Step Heat of HT-Win-06-01b

The diffusivities determined from the step heat ^{3}He and ^{4}He yields for HT-Win-06-01b are always significantly lower than the diffusivities determined from step heat experiments on the same sample in vacuum, and they evolve distinctly from those in vacuum. The Arrhenius plot of the ^{3}He diffusivities initially follows a declining concave-up trend until ~3.5% cumulative ^{3}He yield, then slowly decreases in sub-parallel arrays during prograde and retrograde heating cycles from 3.5-7% cumulative ^{3}He yields (Fig. 2, Fig. 3). The Arrhenius plot of the ^{4}He diffusivities differs in shape: it is concave-down to about 5% cumulative ^{3}He yield, after which it follows a roughly linear array about one log unit below the ^{3}He array (Fig. 2). The ^{3}He diffusivities settled into a linear array which was suitable for linear regression after ~7% of the cumulative ^{3}He yield was lost (Fig. 4b). All remaining diffusivity measurements of the experiment were included in the linear regression, so that diffusivities from ^{3}He loss fractions of about 7-15% were included. The linear regression yields an R^2 of 0.989, an activation energy of 157.1± 5.3kJ/mol, and a closure temperature of 98.1± 8.6°C assuming a cooling rate of 10°C/Ma and considering only the uncertainty in the regression intercept (D_0/a^2).

A delta plot was generated using the intercept and slope parameters from the ^{3}He regression as a reference line, similarly to previous work (Farley, 2018). The ^{3}He apparent diffusivity delta values monotonically decrease

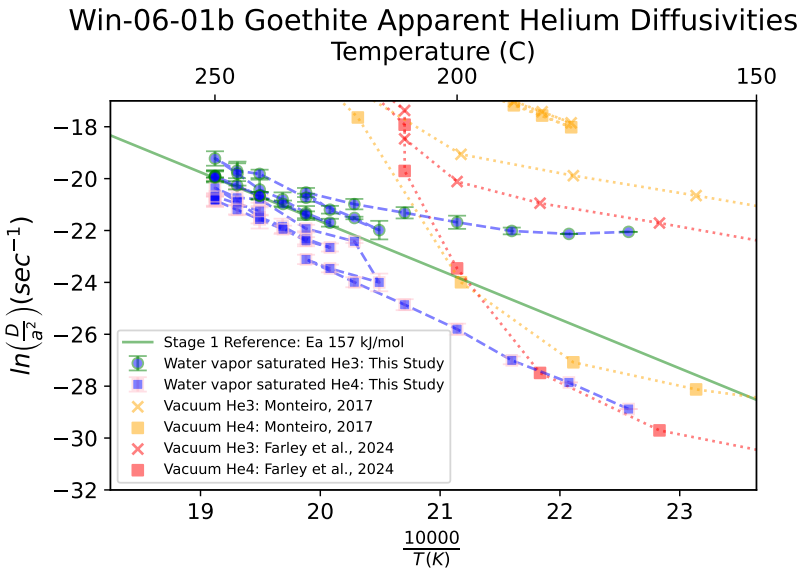


Figure 2: Linear regression of the last 12 measurements of ^{3}He diffusivities in HT-Win-06-01b defining the "Stage 1" reference line. Apparent diffusivities at water vapor saturation are approximately seven natural log units lower than in vacuum. A diffusion domain corresponding to the Stage 1 reference line has a closure temperature of 98.1±8.6°C assuming a cooling rate of 10°C/Ma. Apparent diffusivities calculated from ^{4}He step heats converge to a linear array a single natural log unit lower than ^{3}He. Six steps with ^{3}He yields below 0.1 pccSTP are omitted for clarity.

until a cumulative ^{3}He loss of 7%, after which the delta values are within error of the reference line (Fig. 3).

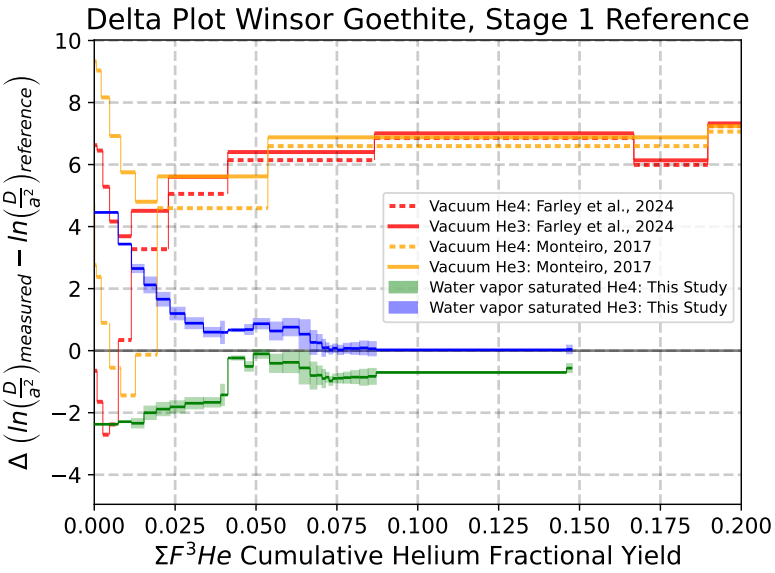


Figure 3: Delta plot similar to those presented in Farley (2018), or r/r0 plots in Lovera et al. (1991) and Farley and Flowers (2012). Six high uncertainty steps with ^{3}He yields below 0.1 pccSTP are omitted for clarity. For visual continuity, the gap left behind is bridged with no uncertainty shown. ^{3}He monotonically decreases until settling on the reference line from 7–15% ^{3}He fractional yield. ^{4}He diffusivities and delta values are generally highly uncertain but initially show an increase in delta values that settles to a slow increase or plateau about one log unit below the ^{3}He reference line by a ^{3}He fractional yield of 5%, which is consistent with diffusive loss of radiogenic ^{4}He in variably retentive domains. Apparent diffusivities for helium in Win-06-01b at water vapor saturation are up to 7 log units lower than the same sample in vacuum.

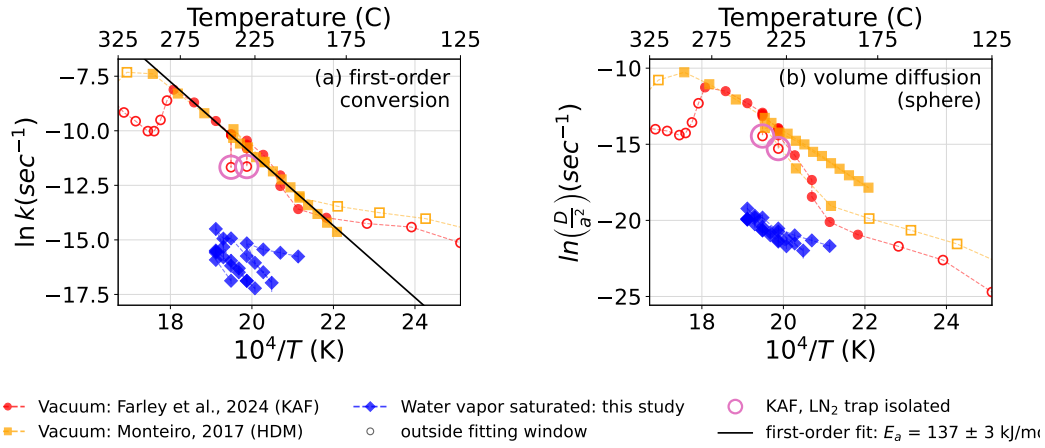


Figure 4: The ^{3}He release of every step of the two vacuum step heats of Win-06-01b (red circles: KAF, Farley et al. 2024; orange squares: HDM, Monteiro 2017) and of the water vapor saturated step heat of this study (blue diamonds), reduced in two ways: (a) as first-order conversion rate constants k_i (Eq. (3), $\phi = 0.101$), and (b) as apparent diffusivities D/a^2 for a sphere (Fechtig and Kalbitzer, 1966). Only the filled points are used for the linear regressions. Large pink circles mark the two KAF steps with the LN_2 trap isolated. Six low-signal steps of the water-saturated experiment are excluded for clarity. The black line is a first-order Arrhenius fit to filled points of both vacuum experiments. These fitted parameters are given in Eq. (4). Below ~180 °C both vacuum experiments depart from the first-order line.

3.2. Kinetic Reanalysis of Vacuum Step Heats

All steps are plotted in Fig. 4, for both the kinetic and diffusion reductions, but the fits use only KAF steps 5–22 (185–280 °C) and HDM steps 8–26 (180–296 °C).

An unweighted least-squares fit of $\ln k_i$ against $1/T_i$ for the 35 steps in the fitting window gives

$$E_a = 137.0 \pm 2.7 \text{ kJ mol}^{-1}, \qquad \ln A = 21.91 \pm 0.66 \tag{4}$$

Fitting KAF and HDM separately (139.3 ± 6.2 and 135.4 ± 2.5 kJ mol^{-1}) or fitting all steps (122.7 kJ mol^{-1}) gives consistent results (§S2).

KAF steps 1–4 and HDM steps 2–7 (below ∼180 °C) release ^{3}He 0.5–3 orders of magnitude faster than the extrapolated conversion rate (Fig. 4). This component amounts to less than 2% of the ^{3}He.

In two KAF steps the trap was isolated and evolved water accumulated in the line. At 230 °C the step rate constant fell to 0.33 times the mean of its neighbors at the same temperature, and at 240 °C to 0.22 times. The rate recovered fully when the trap was restored. We return to these two steps in §4.3.

4. Discussion

4.1. Dehydroxylation Kinetics in Vacuum Step Heats of Win-06-01b

Farley et al. (2024) showed that evolved helium in vacuum step heats of goethite is controlled by the process of dehydroxylation to hematite. Here, we present a reanalysis of the vacuum step heat data of Win-06-01b from KAF and HDM to show that the helium release of Win-06-01b in vacuum

is better described by a model of first-order reaction kinetics for goethite dehydroxylation than by a second order diffusion model.

The kinetic reduction yields simpler, more interpretable Arrhenius plots for the vacuum experiments than the diffusion inversion does. The kinetic inversion collapses the prograde steps at the beginning of the vacuum experiments onto a smooth decay to a single line that the diffusion reduction instead shows as an irreversible increase in diffusivity (Fig. 4). This initial helium release is most easily read from the kinetic inversion as a diffusion-related loss from a broad distribution of small, poorly retentive domains in the goethite that is faster than the rate of dehydroxylation. This is not obvious from the Arrhenius plot produced with the diffusion reduction. This shift in the dominance from diffusive loss to dehydroxylation-related loss offers a clear explanation for the change in behavior between "Stage 1" and "Stage 2" in vacuum step heat experiments described by Farley et al. (2024).

Thermogravimetric studies report somewhat higher apparent activation energies for goethite dehydroxylation (154 ± 15 to 169 ± 8 kJ mol^{-1}; Goss 1987), plausibly because dehydroxylation kinetics depend on particle size, morphology, and aluminum content (Langmuir, 1971; Yapp, 1983; Tardy and Nahon, 1985; Majzlan and Navrotsky, 2003).

4.2. Helium Release under Water-Saturated Conditions Is Diffusive

The goal of this work was to attempt to slow the goethite dehydroxylation process enough to observe thermal helium loss in unchanging goethite. The thermal schedule included several prograde-retrograde heating cycles which together would have completely dehydroxylated goethite in vacuum, using the vacuum kinetics of Eq. (4). Even so, only 14.7% of the ^{3}He was released

in the water-saturated experiment. We argue below that the helium loss in the water-saturated experiment is controlled by diffusion rather than by conversion.

Fig. 4 compares the kinetic and diffusion inversions of the step heat data. These plots show the opposite trend for the vacuum and water vapor saturated experiments. While the vacuum experiments yield simpler, more interpretable arrays with the first-order inversion, the water vapor saturated experiment collapses into a simpler, coherent linear array with the diffusivity reduction. The kinetic inversion of the water-saturated data shows a continuously declining k for any given temperature as the experiment proceeds, forming a zig-zag pattern. This is the expected behavior for a second-order (diffusion-controlled) process for which helium loss slows more strongly than a first-order process as the experiment proceeds.

We consider two alternative mechanisms, recrystallization and fluid-mediated conversion, that might have contributed to the water-saturated release. Poorly crystalline goethite is known recrystallize under hydrothermal conditions at timescales similar to this experiment (Schwertmann et al., 1985). Additionally, we induced a fluid-mediated reaction that transformed synthetic goethite pigment to hematite at similar temperatures and timescales to this experiment (SI, §S6). This mechanism of reaction is not described by the solid-state nucleation model of §4.3. We observed that goethite either entirely reacted or remained unchanged in this experiment, suggesting that the reaction was fast when it occurs. Either process would likely release essentially all of the helium in the material it affects. A gradual reaction would then make helium release track the extent of reaction, a first-order-like behavior that the water-

saturated data do not show (Fig. 4a), and episodic reaction would produce abrupt excess release in individual steps, which is also not observed (Figs. 2 and 3). Their contribution is therefore likely to be small. If either mechanisms contributed, the apparent diffusivities reported here would represent upper bounds for diffusion in goethite.

While it is difficult to prove that decomposition or recrystallization is not playing a role in the results presented in this work, the results presented here point most strongly to a second-order process (likely volume diffusion) as the dominant mechanism of helium loss for Win-06-01b goethite in water-saturated conditions.

4.2.1. Isothermal Discrimination Test at 250 °C

Sequences of isothermal steps can be particularly powerful in discriminating first and second order processes. Diffusion slows quickly as the diffusion front moves inward, depleting the grain rims of helium, whereas the rate of first-order conversion slows less for similar fractional yields. We leveraged this principle to conduct a sensitive test by performing a series of 4 consecutive, long isothermal steps at 250 °C at the end of the water-saturated experiment: three consecutive 5-hour steps and one 170.8-hour step. Using the first three 5 h steps at 250 °C, we invert for both kinetic rate constants and apparent diffusivities, then use each model to project the yield for the 170.8 h step.

Both the first-order and diffusion models yield reproducible results for the three 5-hour calibration steps (Fig. 5), but their projections sharply diverge for the long step. Diffusion projects a final fractional cumulative yield of $F = 0.149$ against an observed $F = 0.146$, a much stronger agreement than

the first-order model prediction of $F = 0.173$, which overpredicts the helium yield for that step by 46% (Fig. 5).

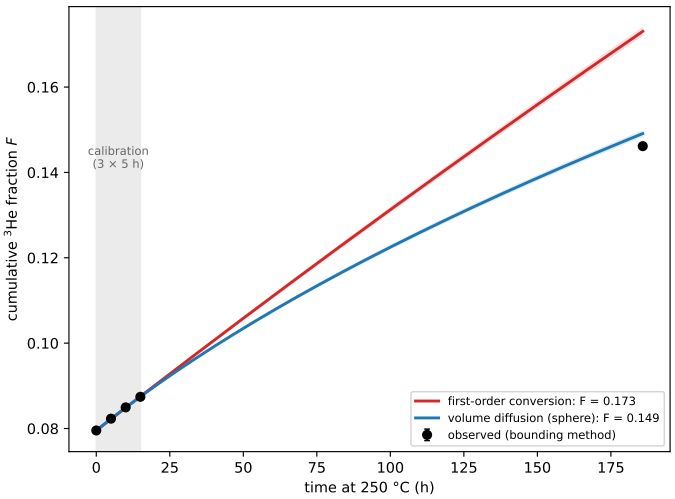


Figure 5: The isothermal 250 °C sequence of the water-saturated experiment. Black markers are measured cumulative ^{3}He fraction. The first-order conversion and spherical volume diffusion models are each calibrated on the three 5 h steps (grey band) and projected through the 170.8 h step.

4.2.2. Consistency with MDD Systematics

All existing helium step heat data on goethites are from naturally occurring polycrystalline aggregates with unclear and potentially widely variable grain sizes (Monteiro, 2017; Farley et al., 2024), which likely could follow MDD systematics (Farley, 2018). In MDD systems, apparent bulk diffusivities of a homogeneously distributed diffusant at a single temperature will decrease as an experiment proceeds. This manifests as a monotonic decrease in a delta or r/r0 plot (Lovera et al., 1991; Farley, 2018). For the activation energy of 157.1 kJ/mol retrieved by regression in this work, the homoge-

neously distributed ^{3}He step heat data show this behavior and monotonically decrease within a delta plot (Fig. 3).

The strong plateau of the water-saturated goethite experiment at $\Delta = 0$ (Fig. 3) and the statistically robust regression over a wide range of fractional cumulative ^{3}He yields (from 7-15% $\Sigma F_{^3\mathrm{He}}$ with an R^2 of 0.989) suggest that the frequency factor (intercept) obtained by this regression may be dominated by a single diffusion domain for $F_c \geq 7\%$ rather than a mix of diffusion domains as is most often observed in the apparent diffusivities for polycrystalline hematite (Farley, 2018).

A multiple diffusion domain inversion of the ^{3}He release (§2.2) tests this claim (Table S5; Fig. 6). A single domain cannot fit the release (Arrhenius $\chi^2_\nu = 27$), and two domains capture the high early diffusivities but leave a poor Arrhenius fit ($\chi^2_\nu = 3.3$). Three domains reduce the yield misfit by more than half ($\chi^2_\nu = 1.08$). With four, the smallest domain splits in two and the fit improves slightly further ($\chi^2_\nu = 1.00$). Every domain added beyond four coincides in size with the dominant domain and leaves the fit unchanged (Fig. 6). We therefore adopt four domains as a complete description of the ^{3}He data:

- 0.57% of the ^{3}He at $r/r_0 = 6.8 \times 10^{-4}$;
- 0.66% at $r/r_0 = 2.9 \times 10^{-3}$;
- 1.5% at $r/r_0 = 0.031$;
- 97.2% at $r/r_0 = 1.12$.

The three small domains, holding 2.8% of the ^{3}He in total, are fully degassed during the experiment and account for the high early diffusivities and the

gradual approach of the delta plot to the Stage 1 line (Fig. 7). The dominant domain releases 12% of its ^{3}He during the experiment and diffuses at 0.80 times the Stage 1 rate. The four-domain model reproduces the Arrhenius array, including its retrograde segments, and the delta plot (Fig. 7). Fits for $N = 1$–5 are shown in the SI (Fig. S6).

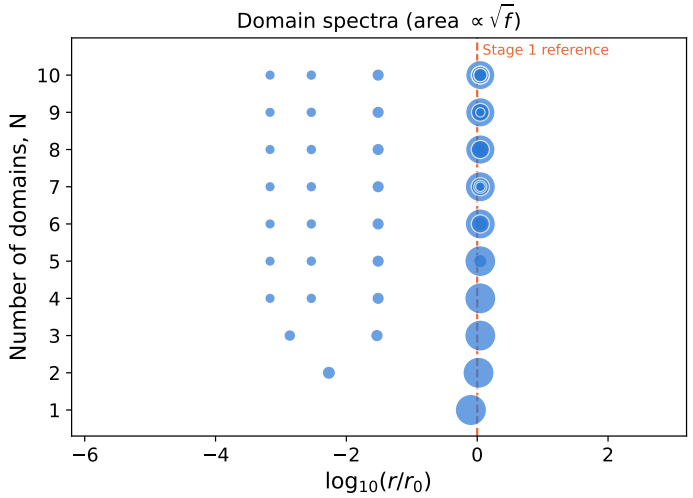


Figure 6: Domain structure for the best-fitting multiple diffusion domain models when 1-10 domains are fit to the ^{3}He release of HT-Win-06-01b. Symbol area scales with the square root of the ^{3}He fraction; the dashed line marks the Stage 1 reference domain. The three small domains are unchanged from $N = 4$ onward; domains added beyond $N = 4$ coincide in size with the dominant domain.

4.3. Why Water Suppresses Dehydroxylation

Under the vacuum kinetics of Eq. (4), the water-saturated heating schedule would have converted the sample completely many times over ($\sum_i k_{\mathrm{vac}}(T_i)\,\Delta t_i \approx 60$).

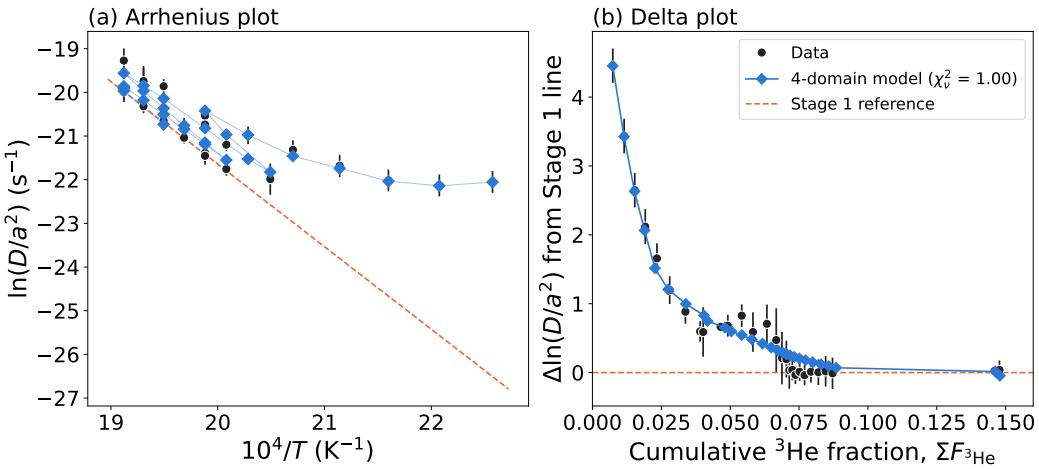


Figure 7: (a) Arrhenius plot and (b) delta plot (departure from the Stage 1 line) of the measured ^{3}He diffusivities of HT-Win-06-01b, with the values implied by the four-domain fit to the ^{3}He yields. Neither plot is fitted directly but is a consequence of the fit to the ^{3}He cumulative and fractional yields. The six low-signal steps are omitted.

A thermodynamic and classical-nucleation model of goethite dehydroxylation accounts for the suppression of the dehydroxylation (SI, §S3–S4). Goethite is unstable relative to hematite and water both in vacuum and at water saturation (SI Fig. S5), but water is a product of the reaction and exerts strong control over the reaction's driving force $\Delta\mu$. In the vacuum of the step-heating line, the water pressure approaches zero and $\Delta\mu$ is unbounded; at water saturation $\Delta\mu$ is only 6–11 kJ per mole Fe_2O_3, 1.6–2.4 times RT (SI Table S4). Because the barrier to nucleating hematite within goethite scales as $\gamma^3/\Delta\mu^2$ (Turnbull and Fisher, 1949), it vanishes in vacuum, consistent with the simple Arrhenius behavior observed in both vacuum experiments, but this nucleation barrier is present at saturation. Classical nucleation theory can be used to predict the ratio of the solid-state conversion rates in

vacuum and at saturation, given the energy of the goethite–hematite interface (γ). Although γ is not known explicitly, any reasonable value predicts no appreciable reaction at saturation. Limiting conversion to 10^{-6} requires a minimum of $\gamma \gtrsim 0.10\ \mathrm{J\,m^{-2}}$, a factor of 2.4 below the energy of twin boundaries within hematite formed by dehydrating goethite (Saito et al., 2016). An interface between two different phases is unlikely to undercut a stable interface within a single phase (SI, §S4). This explanation applies to solid-state (topotactic) conversion and does not pertain to fluid-mediated reaction (§4.2).

4.4. High Helium Retentivity of Win-06-01b-HT

The apparent diffusivities of the water-saturated experiment are approximately seven natural log units lower than those obtained from the same sample in vacuum by Monteiro (2017) and Farley et al. (2024) (Fig. 4b, Fig. 2, Fig. 3). This high helium retentivity is explored through two lenses: closure temperature of cooling samples (Dodson, 1973) and isothermal diffusion (Wolf et al., 1998).

If the slope and intercept on the regression of ^{3}He apparent diffusivities from Fig. 2 are taken to represent a real diffusion domain, then it has a closure temperature of 98.1± 8.6°C assuming a cooling rate of 10°C/Ma and considering uncertainty only in the intercept D_0/a^2 (Dodson, 1973). The multiple diffusion domain inversion (§4.2.2) indicates that the dominant domain, which holds 97% of the ^{3}He, diffuses at 0.80 times the rate of the Stage 1 line. Its closure temperature is correspondingly about 1.6 °C higher, well within the closure temperature's uncertainty, so the Stage 1 parameters are used throughout this section.

Goethite is most often formed near Earth's surface and thus U-Th/He geochronology is most typically applied to date surface processes (Heim et al., 2006; Yapp and Shuster, 2011; Vasconcelos et al., 2013; Danišík et al., 2013; Monteiro et al., 2014; Hofmann et al., 2017; Allard et al., 2018). For crystals of goethite grown and collected near the surface, the concept of a closure temperature is poorly applicable, as it necessarily assumes a cooling rate (Dodson, 1973). In such cases, a better measure of retentivity is an analysis of isothermal retentivity over time (Wolf et al., 1998), which is commonly expressed in terms of t'/t, where t' is the helium age and t is the isothermal holding time (often the 'real age'). For an isothermal holding time of 100 Ma at 40°C, a diffusive domain with the parameters obtained from the regression of ^{3}He apparent diffusivities of Win-06-01b-HT yields t'/t = 0.968, where t' and t are helium age and isothermal holding time, respectively. At 25°C, this domain is nearly quantitatively retentive of helium even at billion-year time scales (t'/t $\approx$ 1).

This study of a single sample has no bearing on potential complexities related to aluminum content, radiation damage, or crystal defects on bulk diffusion (Bassal et al., 2022), nor does it permit estimation of the diffusivity at infinite temperature (D0) which would require experiments on controlled grain sizes.

4.5. Profiling Radiogenic Helium Loss with $^{4}He/^{3}He$ Ratios of Win-06-01b-HT

Bulk-normalized ^{4}He/^{3}He ratios in step heats of irradiated samples measure the rounding of concentration profiles of radiogenic helium by natural helium loss, if the loss mechanism in nature and in the laboratory share the

same geometry (Shuster and Farley, 2004; Farley et al., 2024). Aside from the first experimental step, bulk-normalized ^{4}He/^{3}He ratios for all experiments begin at vanishingly small values and steeply increase at first, with the rate of increase slowing as the experiments proceed (Fig. 8). This behavior is consistent with complete then partial loss of radiogenic helium in nature in the least retentive domains and is observed in all step heat experiments on irradiated goethite (Shuster et al., 2005; Vasconcelos et al., 2013; Farley et al., 2024). These least retentive domains could represent fine goethite grains, or they could represent interstitial helium trapped between crystallite grain boundaries.

The water vapor saturated experiment records lower bulk-normalized ^{4}He/^{3}He ratios than the vacuum experiments at any given $\Sigma F_{^3\mathrm{He}}$ for the duration of the experiment. This suggests that the geometry of helium loss differs between the water vapor saturated and vacuum experiments. If this is the case, then the ^{4}He/^{3}He spectra of goethite measured in vacuum may not straightforwardly encode the thermal history of the sample in the same way that it does in phases such as hematite or apatite (Farley and McKeon, 2015; Shuster and Farley, 2004). More ^{4}He/^{3}He spectra of goethite at water vapor saturation are needed to investigate this hypothesis further.

These ratios also bear on the natural thermal history of Win-06-01b. Throughout the range of $\Sigma F_{^3\mathrm{He}}$ that defines the Stage 1 reference line (7–15%), step ^{4}He/^{3}He ratios remain at 0.67–0.82 of the bulk value (Fig. 8). Radiogenic ^{4}He is depleted relative to the uniformly distributed ^{3}He in this domain, so even this retentive domain lost helium in nature. A domain with the Stage 1 parameters is nearly quantitatively retentive at Earth-surface

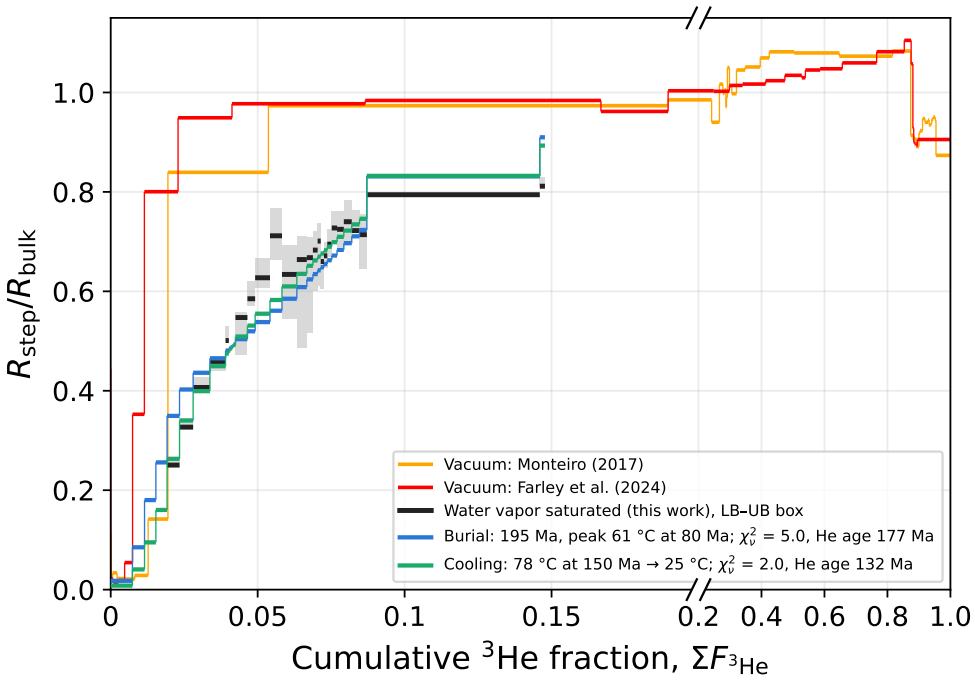


Figure 8: Bulk-normalized ^{4}He/^{3}He ratios for Win-06-01b goethite step heats against cumulative fractional ^{3}He yield. The water vapor saturated experiment of this study is black, with uncertainties shown in grey. The vacuum experiments of Monteiro (2017) and Farley et al. (2024) are orange and red respectively. The best age-consistent thermal-history predictions (shown as stars in Fig. 9) are shown here in blue for the burial history with crystallization at 195 Ma, and in freen for the steady cooling history (§4.6). The water vapor saturated experiment shows lower bulk-normalized ^{4}He/^{3}He ratios at a given $\Sigma F_{^3\mathrm{He}}$ for the duration of the experiment, suggesting that ^{4}He/^{3}He spectra are dependent on the partial pressure of water and that the geometry of goethite dehydroxylation differs from the geometry of helium diffusion through goethite. Steps of the water vapor saturated experiment with less than 0.1 pccSTP of ^{3}He are omitted. The horizontal axis changes scale at $\Sigma F_{^3\mathrm{He}} = 0.2$ (double slash) to emphasize the portion of the plot where water vapor saturated data exist.

temperatures (§4.4), so this loss is difficult to attribute to thermally activated diffusion at surface conditions. Either Win-06-01b experienced temperatures well above surface conditions (perhaps it grew in a hydrothermal setting or experienced burial heating), or non-thermal processes such as fluid-mediated recrystallization caused partial helium loss in nature. Section 4.6 tests whether thermally activated diffusion alone, within the domain structure inferred from the ^{3}He release, can reproduce the measured spectrum.

4.6. Thermal History of Win-06-01b

To test whether thermally activated diffusion alone can reproduce the measured ^{4}He/^{3}He spectrum, we explore a subspace of geologically reasonable, yet simple thermal paths. With the four-domain structure of §4.2.2 held fixed, we fit steady-cooling and burial-and-exhumation thermal histories to the ^{4}He/^{3}He spectrum of HT-Win-06-01b (§2.2). Previous studies constrain our exploration of the burial-and-exhumation paths. Gibson and Stüwe (2000); Mitchell et al. (1998) interpret regional cooling from roughly 50–70 °C after the mid-Cretaceous from their apatite fission-track studies. Mitchell et al. (1998) additionally interpret, from limited ∼1.5 km thick sediment accumulation in Triassic intramontane basins, that parts of the ranges were near the surface in the early Mesozoic. Table S6 gives the best fit of each family, Fig. 8 compares the best steady-cooling and burial predictions with the spectrum, and Fig. 9 shows their χ^2 maps on a common color scale.

Steady cooling. Linear cooling at a rate of ∼0.3 °C Myr^{-1} from 78 °C at 150 Ma ($\chi^2_\nu = 2.00$) to 25 °C today fits the spectrum best. Such paths pass through ∼55 °C at 90 Ma, within constrains set by the apatite fission-

track (Gibson and Stüwe, 2000; Mitchell et al., 1998). The spectrum alone without the age constraint prefers somewhat younger, hotter paths: the unconstrained minimum ($\chi^2_\nu = 1.68$; 89 °C at 80 Ma) predicts a He age of 68 Ma (Fig. 9b).

Burial and exhumation. Burial paths fit the spectrum less well ($\chi^2_\nu = 5.0$–6.5), but they agree with the independent regional constraints of Gibson and Stüwe (2000) and Mitchell et al. (1998). For crystallization at 180 and 195 Ma, the best age-consistent paths peak at ∼60 °C at 70 and 80 Ma (Fig. 9a). These peaks fall in or just after the apatite fission-track window, within the 50–70 °C range of Gibson and Stüwe (2000) and at the ≤60 °C limit reported by Mitchell et al. (1998). For crystallization at 210 Ma, only hotter and earlier peaks (70–74 °C at 105–130 Ma) give an age within the measured range, and they fit worse ($\chi^2_\nu = 6.5$).

Isothermal histories, which constrain only the product of diffusivity and time, cannot reproduce the shape of the spectrum and are not considered further (Table S6).

Where the families differ. The steady-cooling and burial predictions differ mainly in the earliest steps ($\Sigma F_{^3\mathrm{He}} \lesssim 0.035$; Fig. 8), which sample the outer rim of the dominant domain. On the burial path the sample sits at 20 °C for the last 50 Myr, long enough for radiogenic ^{4}He to re-accumulate in the rim, so the early ratios are overpredicted. Steady cooling keeps the rim depleted until geologically recent times, with diffusive loss declining monotonically toward the present. Elsewhere both leave the same residual pattern: the underpredict $^4\mathrm{He}/^3\mathrm{He}$ ratios between $\Sigma F_{^3\mathrm{He}} = 0.04$ and 0.06, and overpredict them in for the two longest high-temperature steps. These fits do not weight

steps differently by their fractional yields except indirectly in that longer steps have tighter uncertainties and thus incur more penalty for a given misfit. The trend of the residuals at the end of the experiment suggest that the best fitting thermal history would be different if the experiment continued further and more helium was degassed.

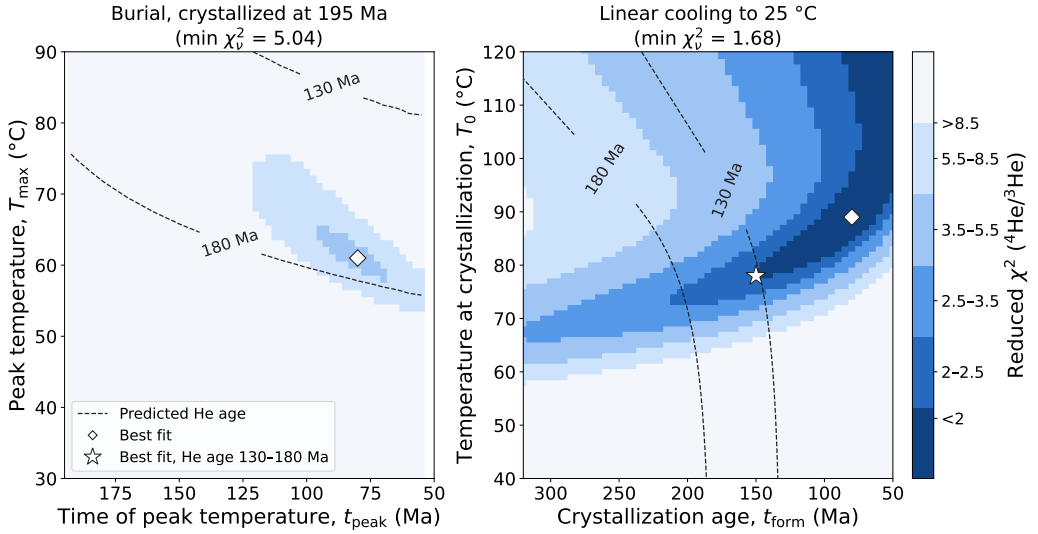


Figure 9: Reduced χ^2 of the ^{4}He/^{3}He spectrum for (a) burial paths with crystallization at 195 Ma, as a function of the time and magnitude of peak temperature, and (b) steady cooling to 25 °C, as a function of crystallization age and initial temperature. Both panels share one color scale, anchored to the overall minimum. Dashed lines show the predicted bulk He age at 130 and 180 Ma. A diamond is placed at the unconstrained best fit for the linear cooling path, and stars locate the best fit with a He age within the 130–180 Ma bounds set by Waltenberg (2012).

Consistency with diffusive loss. These two families of thermal paths encode very different geologic origins. We note that any number of arbitrarily complex thermal paths can be evaluated against the data with this workflow, but

a thorough search for viable thermal paths is beyond the scope of this study, and is an ill-posed exercise for a single sample, for which many potential histories could fit the data non-uniquely. The main purpose of this exercise is instead to demonstrate that the $^4He/^3He$ data we present here are consistent with diffusive loss of radiogenic helium in nature. We do not claim to have found the unique thermal history of the sample, but have demonstrated that plausible thermal histories reproduce our results reasonably well. This result suggests that proton-irradiated goethite may be a viable target for $^4He/^3He$ thermochronometry as described by Shuster and Farley (2004) and applied to polycrystalline samples by Farley and Flowers (2012), especially when multiple samples are available to constrain a region's thermal history.

5. Conclusions

We applied the silica-glass ampoule method of Thyer et al. (2026) to conduct the first helium step heat experiment on goethite under water-saturated conditions, using proton-irradiated goethite Win-06-01b, which was previously been step heated in vacuum. The main findings are:

1. Polycrystalline goethite heated with water does not develop the contraction cracks that form in vacuum, and the water-saturated helium release differs fundamentally from release in vacuum. Published vacuum step heats of Win-06-01b follow first-order conversion to hematite, whereas the water-saturated release shows the history and time dependence of diffusion. These results are consistent with the interpretation that goethite dehydroxylation was sufficiently kinetically-inhibited to observe helium diffusion through unchanging goethite.

2. Apparent diffusivities at water saturation are approximately seven natural log units lower than those of the same sample in vacuum. Beyond 7% cumulative ^{3}He release they define a linear Arrhenius array with an activation energy of 157.1 ± 5.3 kJ/mol, and the delta plot follows multiple diffusion domain systematics. A four-domain model places 97% of the ^{3}He in a single domain.
3. Extrapolated to 25 °C, this majority domain is essentially quantitatively retentive of helium over billion-year timescales, supporting the use of goethite (U–Th)/He dates to date surface processes.
4. The ^{4}He/^{3}He spectrum shows that even this retentive domain lost radiogenic helium in nature, and thermally activated diffusion can account for that loss. Steady cooling from 76–80 °C at 150–175 Ma fits the spectrum best, favoring crystallization from warm, possibly hydrothermal, fluids. Burial of surface-formed goethite to a Late Cretaceous peak of 58–65 °C, consistent with regional apatite fission-track constraints, remains a viable alternative.
5. The water-saturated ^{4}He/^{3}He spectrum is distinct from the vacuum spectra of the same sample, suggesting that dehydroxylation and diffusion may degas goethite with different geometries. Vacuum ^{4}He/^{3}He spectra of goethite may therefore not encode thermal histories as directly as they do for minerals that do not decompose during vacuum step heat experiments.
6. A thermodynamic and kinetic analysis explains why water preserves goethite on experimental timescales. The driving force for dehydroxylation is unbounded at the near-zero water pressure of a vacuum line,

but is only 6–11 kJ at saturation, which raises the barrier to topotactic hematite nucleation enough to suppress conversion over experimental timescales.

Fluid-mediated conversion and recrystallization, which transformed synthetic goethite pigment under similar conditions, would likely degas the affected material completely. These data are most consistent with a second-order control, limiting the contribution of fluid-mediated conversion and recrystallization to be small. If either contributed, the diffusivities reported here are upper bounds and the retentivity a lower bound. Separating the effects of crystallite size, chemical composition, and radiation damage on helium retention will require further water-saturated step heats of irradiated goethite of controlled grain size, which this approach makes straightforward.

Data Availability

The step heat data, permeation-lag bounds, and model outputs supporting this study are archived on Zenodo (Thyer, 2026a): https://doi.org/10.5281/zenodo.22113166.

CRediT authorship contribution statement

Ben J. Thyer: Methodology, Investigation, Formal analysis, Visualization, Writing – original draft. **Jonathan Treffkorn:** Methodology, Investigation. **Kenneth A. Farley:** Conceptualization, Resources, Supervision, Funding acquisition, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by National Science Foundation (NSF) grant EAR 1945974 to K.A. Farley. Hevelyn Monteiro is thanked for her mentorship and resources provided to BJT in support of the SEM work presented here. This manuscript benefitted greatly from conversations with Ed Stolper and Paul Asimow.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Claude (Anthropic) to assist with software programming and with the editing and LaTeX typesetting of the manuscript. After using this tool, the authors reviewed and edited all content and take full responsibility for the content of the publication.

References

Allard, T., Gautheron, C., Bressan Riffel, S., Balan, E., Soares, B.F., Pinna-Jamme, R., Derycke, A., Morin, G., Bueno, G.T., do Nascimento, N., 2018. Combined dating of goethites and kaolinites from ferruginous duricrusts: Deciphering the late neogene erosion history of central amazonia. Chemical Geology 479, 136–150. doi:10.1016/j.chemgeo.2018.01.004.

Ansart, C., Quantin, C., Calmels, D., Allard, T., Roig, J.Y., Coueffe, R., Heller, B., Pinna-Jamme, R., Nouet, J., Reguer, S., Vantelon, D., Gautheron, C., 2022. (U-Th)/He Geochronology Constraints on Lateritic Duricrust Formation on the Guiana Shield. Frontiers in Earth Science 10. doi:10.3389/feart.2022.888993.

Bassal, F., Heller, B., Roques, J., Balout, H., Tassan-Got, L., Allard, T., Gautheron, C., 2022. Revealing the radiation damage and al-content impacts on he diffusion in goethite. Chemical Geology 611, 121118. doi:10.1016/j.chemgeo.2022.121118.

Cherniak, D.J., Watson, E.B., 2011. Helium diffusion in rutile and titanite, and consideration of the origin and implications of diffusional anisotropy. Chemical Geology 288, 149–161.

Danišík, M., Evans, N.J., Ramanaidou, E.R., McDonald, B.J., Mayers, C., McInnes, B.I.A., 2013. (u-th)/he chronology of the robe river channel iron deposits, hamersley province, western australia. Chemical Geology 354, 150–162. doi:10.1016/j.chemgeo.2013.06.012.

Deng, X.D., Li, J.W., Shuster, D.L., 2017. Late mio-pliocene chemical weathering of the yulong porphyry cu deposit in the eastern tibetan plateau constrained by goethite (u-th)/he dating: Implication for asian summer monsoon. Earth and Planetary Science Letters 472, 289–298.

Dodson, M.H., 1973. Closure temperature in cooling geochronological and petrological systems. Contributions to Mineralogy and Petrology 40, 259–274.

Farley, K.A., 2000. Helium diffusion from apatite: General behavior as illustrated by durango fluorapatite. Journal of Geophysical Research: Solid Earth 105, 2903–2914.

Farley, K.A., 2007. He diffusion systematics in minerals: Evidence from synthetic monazite and zircon structure phosphates. Geochimica et Cosmochimica Acta 71, 4015–4024.

Farley, K.A., 2018. Helium diffusion parameters of hematite from a single-diffusion-domain crystal. Geochimica et Cosmochimica Acta 231, 117–129.

Farley, K.A., Flowers, R.M., 2012. (U–Th)/Ne and multidomain (U–Th)/He systematics of a hydrothermal hematite from eastern Grand Canyon. Earth and Planetary Science Letters 359-360, 131–140.

Farley, K.A., McKeon, R., 2015. Radiometric dating and temperature history of banded iron formation-associated hematite, Gogebic iron range, Michigan, USA. Geology 43, 1083–1086.

Farley, K.A., Monteiro, H.B., Vasconcelos, P.M., Waltenberg, K., 2024. Dehydration of goethite during vacuum step-heating and implications for he retentivity characterization. Chemical Geology 663, 122254.

Farley, K.A., Reiners, P.W., Nenow, V., 1999. An apparatus for high-precision helium diffusion measurements from minerals. Analytical Chemistry 71, 2059–2061.

Fechtig, H., Kalbitzer, S., 1966. The diffusion of argon in potassium-bearing solids, in: Schaeffer, O.A., Zähringer, J. (Eds.), Potassium Argon Dating. Springer, Berlin, Heidelberg, pp. 68–107.

Gibson, H.J., Stüwe, K., 2000. Multiphase cooling and exhumation of the southern Adelaide Fold Belt: constraints from apatite fission track data. Basin Research 12, 31–45. doi:10.1046/j.1365-2117.2000.00110.x.

Goss, C.J., 1987. The kinetics and reaction mechanism of the goethite to hematite transformation. Mineralogical Magazine 51, 437–451.

Guenthner, W.R., Reiners, P.W., Ketcham, R.A., Nasdala, L., Giester, G., 2013. Helium diffusion in natural zircon: Radiation damage, anisotropy, and the interpretation of zircon (u-th)/he thermochronology. American Journal of Science 313, 145–198.

Guo, H., Barnard, A., 2011. Thermodynamic modelling of nanomorphologies of hematite and goethite. Journal of Materials Chemistry 21, 11566–11577.

Haar, L., Gallagher, J.S., Kell, G.S., 1984. NBS/NRC Steam Tables: Thermodynamic and Transport Properties and Computer Programs for Vapor and Liquid States of Water in SI Units.

Heim, J.A., Vasconcelos, P.M., Shuster, D.L., Farley, K.A., Broadbent, G., 2006. Dating paleochannel iron ore by (u-th)/he analysis of supergene goethite, Hamersley province, Australia. Geology 34, 173–176.

Hofmann, F., Reichenbacher, B., Farley, K.A., 2017. Evidence for >5 ma paleo-exposure of an eocene–miocene paleosol of the bohnerz formation, switzerland. Earth and Planetary Science Letters 465, 168–175.

Langmuir, D., 1971. Particle size effect on the reaction goethite = hematite + water. American Journal of Science 271, 147–156. URL: https://ajso

nline.org/article/59465-particle-size-effect-on-the-reaction-goethite-hematite-water, doi:10.2475/ajs.271.2.147.

Lovera, O.M., Richter, F.M., Harrison, T.M., 1991. Diffusion domains determined by 39ar released during step heating. Journal of Geophysical Research: Solid Earth 96, 2057–2069.

Majzlan, J., Grevel, K.D., Navrotsky, A., 2003. Thermodynamics of Fe oxides: Part II. enthalpies of formation and relative stability of goethite (α-FeOOH), lepidocrocite (γ-FeOOH), and maghemite (γ-Fe2O3). American Mineralogist 88, 855–859.

Majzlan, J., Navrotsky, A., 2003. Thermodynamics of the goethite-diaspore solid solution. European Journal of Mineralogy 15, 495–501. URL: https://doi.org/10.1127/0935-1221/2003/0015-0495, doi:10.1127/0935-1221/2003/0015-0495.

Mitchell, M.M., Kohn, B.P., Foster, D.A., 1998. Post-orogenic cooling history of eastern South Australia from apatite FT thermochronology, in: van den Haute, P., de Corte, F. (Eds.), Advances in Fission-Track Geochronology: A selection of papers presented at the International Workshop on Fission-Track Dating, Ghent, Belgium, 1996. Springer Netherlands, Dordrecht, pp. 207–224. doi:10.1007/978-94-015-9133-1_13.

Monteiro, H., 2017. Paleoenvironmental Evolution of Continental Landscapes through Combined High-Resolution Geochronology and δ^{18}O Ion Microprobe Analysis of Goethite. Ph.D. thesis. The University of Queensland. doi:10.14264/uql.2018.6.

Monteiro, H.S., Vasconcelos, P.M., Farley, K.A., Spier, C.A., Mello, C.L., 2014. (u-th)/he geochronology of goethite and the origin and evolution of cangas. Geochimica et Cosmochimica Acta 131, 267–289.

Navrotsky, A., Mazeina, L., Majzlan, J., 2008. Size-driven structural and thermodynamic complexity in iron oxides. Science 319, 1635–1638.

Reiners, P.W., Farley, K.A., 1999. Helium diffusion and (u-th)/he thermochronometry of titanite. Geochimica et Cosmochimica Acta 63, 3845–3859.

Saito, G., Kunisada, Y., Nomura, T., Sakaguchi, N., Akiyama, T., 2016. Twin formation in hematite during dehydration of goethite. Physics and Chemistry of Minerals 43, 749–757. doi:10.1007/s00269-016-0831-8.

Schwertmann, U., Cambier, P., Murad, E., 1985. Properties of goethites of varying crystallinity. Clays and Clay Minerals 33, 369–378.

Shuster, D.L., Farley, K.A., 2004. 4he/3he thermochronometry. Earth and Planetary Science Letters 217, 1–17. doi:10.1016/S0012-821X(03)00595-8.

Shuster, D.L., Farley, K.A., Sisterson, J.M., Burnett, D.S., 2004. Quantifying the diffusion kinetics and spatial distributions of radiogenic 4he in minerals containing proton-induced 3he. Earth and Planetary Science Letters 217, 19–32.

Shuster, D.L., Flowers, R.M., Farley, K.A., 2006. The influence of natural radiation damage on helium diffusion kinetics in apatite. Earth and Planetary Science Letters 249, 148–161.

Shuster, D.L., Vasconcelos, P.M., Heim, J.A., Farley, K.A., 2005. Weathering geochronology by (u-th)/he dating of goethite. Geochimica et Cosmochimica Acta 69, 659–673.

Tardy, Y., Nahon, D., 1985. Geochemistry of laterites, stability of Al-goethite, Al-hematite, and Fe^{3+}-kaolinite in bauxites and ferricretes; an approach to the mechanism of concretion formation. American Journal of Science 285, 865–903. URL: https://ajsonline.org/article/60331-geochemistry-of-laterites-stability-of-al-goethite-al-hematite-and-fe-super-3-kaolinite-in-bauxites-and-ferricretes-an-approach-to-the-mechani, doi:10.2475/ajs.285.10.865.

Thyer, B., 2026a. Datasets for "Hydrothermal Helium Step Heats Reveal High Helium Retentivity and Multiple Diffusion Domain Systematics in Goethite". Zenodo dataset. URL: https://doi.org/10.5281/zenodo.22113166, doi:10.5281/zenodo.22113166.

Thyer, B.J., 2026b. Recipes for Rare Metal Ores: Magmas, Metasomatism, and Meteoric Mobilization via Sulfide/Goethite (U-Th)/He Geochronology and ML-Accelerated Thermodynamic Modeling. Ph.D. thesis. California Institute of Technology. Pasadena, CA, USA.

Thyer, B.J., Treffkorn, J., Farley, K.A., 2026. A simple method for helium step heat experiments at high water or sulfur fugacity. Geochronology Discussions In preparation.

Turnbull, D., Fisher, J.C., 1949. Rate of nucleation in condensed systems. The Journal of Chemical Physics 17, 71–73. URL: https://pubs.aip.o

rg/jcp/article/17/1/71/200648/Rate-of-Nucleation-in-Condensed-Systems, doi:10.1063/1.1747055.

Vasconcelos, P.M., Heim, J.A., Farley, K.A., Monteiro, H., Waltenberg, K., 2013. 40ar/39ar and (u-th)/he–4he/3he geochronology of landscape evolution and channel iron deposit genesis at Lynn Peak, Western Australia. Geochimica et Cosmochimica Acta 117, 283–312. doi:10.1016/j.gca.2013.03.037.

Waltenberg, K.M., 2012. Mineral physics and crystal chemistry of minerals suitable for weathering geochronology: Implications to 40ar/39ar and (u-th)/he geochronology. PhD thesis; institution not confirmed in the source draft.

Wolf, R.A., Farley, K.A., Kass, D.M., 1998. Modeling of the temperature sensitivity of the apatite (u-th)/he thermochronometer. Chemical Geology 148, 105–114.

Yapp, C.J., 1983. Effects of AlOOH-FeOOH solid solution on goethite-hematite equilibrium. Clays and Clay Minerals 31, 239–240. URL: https://www.cambridge.org/core/journals/clays-and-clay-minerals/article/abs/effects-of-aloohfeooh-solid-solution-on-goethitehematite-equilibrium/C3129756B1F09315D108151298352608, doi:10.1346/CCMN.1983.0310312.

Yapp, C.J., Shuster, D.L., 2011. Environmental memory and a possible seasonal bias in the stable isotope composition of (u-th)/he-dated goethite

from the canadian arctic. Geochimica et Cosmochimica Acta 75, 4194–4215.